THE DIFFERENCE OF THE FIRST RECURRENCE MATRICES AS A CAUCHY INTEGRAL

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Allowed standard prerequisites (editorial).

- The Riemann-Hilbert problem for the two weights and its jump matrices. same article, the displays at source lines 674 and 745 and the relation at source line 761. Fixes the jump matrices of the two solutions and the relations between them, which are what the auxiliary matrix of the proof is built from.
- The Fokas-Its-Kitaev problem for the companion weight and its solution. same article, the display at source line 745. Provides the model solution with respect to which the behaviour of the auxiliary matrix is controlled.
- The Sokhotski-Plemelj formula for a matrix valued function with a jump on a contour. standard Riemann-Hilbert material used at proof lines 983-989. Recovers the auxiliary matrix from its jump and its normalization at infinity, which is the step that produces the integral formula.

Source reference. Original equation internal: sources/sigma-2024-004-recurrence-coefficients-for-orthogonal-polyn/pinned-original-source.tex, source lines 674–678. The counted proof refers to that the author's displayed jump matrix of the first solution; it is the author's own earlier display, definition, figure or statement in the same article and the whole argument of the counted span is retained verbatim.

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The counted proof refers to that the author's displayed model solution of the companion Riemann-Hilbert problem; it is the author's own earlier display, definition, figure or statement in the same article and the whole argument of the counted span is retained verbatim.

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Source reference. Original equation internal: sources/sigma-2024-004-recurrence-coefficients-for-orthogonal-polyn/pinned-original-source.tex, source lines 541–545. The counted proof refers to that the author's displayed decay estimate for the exponential factor; it is the author's own earlier display, definition, figure or statement in the same article and the whole argument of the counted span is retained verbatim.

Proposition 3.5 ([1, Proposition 4.9])**.** *The following formula holds:*

$$Q_1^{(n)} - \widehat{Q}_1^{(n)} = -\frac{1}{2\pi \mathrm{i}} \int_\Sigma Q_-^{(n)}(s)\big(v^{(n)}(s) - \widehat{v}^{(n)}(s)\big)\big[\widehat{Q}_-^{(n)}(s)\big]^{-1}\mathrm{d}s. \tag{3.1}$$

Proof. In the following, we drop the subscript (n) for better readability, i.e., write Q_1 for $Q_1^{(n)}$ and so on. Let us define the matrix-valued function $X(z) = Q(z)\big[\widehat{Q}(z)\big]^{-1}$ for $z \in \mathbb{C} \setminus \Sigma$. Note that

$$X(z) = I + \frac{Q_1 - \widehat{Q}_1}{z} + O\left(\frac{1}{z^2}\right) \qquad \text{as } z \to \infty. \tag{3.2}$$

Moreover, X satisfies the jump condition

$$X_+(s) = X_-(s) v_X(s), \qquad s \in \Sigma,$$

where $v_X = \widehat{Q}_- v \widehat{v}^{-1} \widehat{Q}_-^{-1}$. We claim that $X_\pm \in L^1(\Sigma)$. To see this let us first introduce the analog of T for the weight function $\widehat{w}$:

$$\widehat{T}(z) = \left(\frac{2}{\widehat{F}_\infty}\right)^{n\sigma_3} \widehat{Y}(z) \left(\frac{\widehat{F}(z)}{\phi^n(z)}\right)^{\sigma_3}.$$

Here $\hat{Y}$ is the solution to the Fokas–Its–Kitaev problem for the weight function $\widehat{w}$. Then for $z \in \Omega_0$ (cf. Figure internal), we have

$$X(z) = T(z)\big[\widehat{T}(z)\big]^{-1} = O(1), \qquad z \in \Omega_0, \quad z \to -1,$$

where we have used (internal) and its analog (see [2, Section 2])

$$\widehat{T}(z) = \begin{pmatrix} O\big(|z+1|^{1/2}\big) & O\big(|z+1|^{-1/2}\big) \\ O\big(|z+1|^{1/2}\big) & O\big(|z+1|^{-1/2}\big) \end{pmatrix}, \qquad z \to -1. \tag{3.3}$$

Note that the analog of (internal) remains valid

$$\widehat{Q}(z) = \begin{cases} \widehat{T}(z), & z \in \Omega_0, \\ \widehat{T}(z) \begin{pmatrix} 1 & 0 \\ -\frac{\widehat{F}^2}{\widehat{w}}(z)\phi^{-2n}(z) & 1 \end{pmatrix}, & z \in \Omega_1, \\ \widehat{T}(z) \begin{pmatrix} 1 & 0 \\ \frac{\widehat{F}^2}{\widehat{w}}(z)\phi^{-2n}(z) & 1 \end{pmatrix}, & z \in \Omega_2. \end{cases}$$

Thus for $z \in \Omega_1 \cup \Omega_2$, we have

$$X(z) = T(z) \begin{pmatrix} 1 & 0 \\ \mp\big(\frac{F^2}{w}(z) - \frac{\widehat{F}^2}{\widehat{w}}(z)\big)\phi^{-2n}(z) & 1 \end{pmatrix} \big[\widehat{T}(z)\big]^{-1} = O(1),$$
$$z \in \Omega_1 \cup \Omega_2, \qquad z \to -1,$$

where the $+$, resp. $-$ sign refer to Ω_1 and Ω_2. Apart from (internal) and (3.3), we have used in addition (internal). As X can have only logarithmic singularities for $z \to 1$ and otherwise the limits to the contour are analytic, it follows that $X_\pm \in L^1(\Sigma)$.

Next, we can use the Sokhotski–Plemelj theorem to conclude that X can be represented as

$$\begin{aligned} X(z) &= I + \frac{1}{2\pi \mathrm{i}} \int_\Sigma \frac{X_+(s) - X_-(s)}{s - z} \mathrm{d}s \\ &= I + \frac{1}{2\pi \mathrm{i}} \int_\Sigma \frac{Q_-(s)\big(v(s)[\widehat{v}(s)]^{-1} - I\big)\big[\widehat{Q}_-(s)\big]^{-1}}{s - z} \mathrm{d}s \\ &= I + \frac{1}{2\pi \mathrm{i}} \int_\Sigma \frac{Q_-(s)(v(s) - \widehat{v}(s))[\widehat{Q}_-(s)]^{-1}}{s - z} \mathrm{d}s, \end{aligned}$$

where we made use of the particular form of the jump matrices v and $\widehat{v}$. This representation together with (3.2) implies (3.1). □

References

[1] Conway T.O., Deift P., Asymptotics of polynomials orthogonal with respect to a logarithmic weight, *SIGMA* **14** (2018), 056, 66 pages, arXiv:1711.01590.

[2] Kuijlaars A.B.J., McLaughlin K.T.-R., Van Assche W., Vanlessen M., The Riemann–Hilbert approach to strong asymptotics for orthogonal polynomials on $[-1, 1]$, *Adv. Math.* **188** (2004), 337–398, arXiv:math.CA/0111252.